

Topic 5: Energy for Biological Processes

This topic builds on knowledge of carbohydrates and enzymes. It considers the sources of energy in living organisms and how energy transfers take place. Details of the stages in respiration and photosynthesis, the roles of co-enzymes and electrons along with the uses of intermediates are included. Practical skills are developed in the investigations of photosynthetic pigments and the factors that affect rates of respiration and photosynthesis.

Opportunities for developing mathematical skills within this topic include: translating information between graphical, numerical and algebraic forms; plotting variables from experimental data; understanding that $y = mx + c$ represents a linear relationship; determining the intercept of a graph; calculating rate of change from a graph showing a linear relationship; determining the intercept of a graph; drawing and using the slope of a tangent to a curve as a measure of rate of change. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:

5.1 Aerobic respiration

- i Know that cellular respiration yields ATP which is used as a source of energy for metabolic reactions, and the process also generates heat.
- ii Know the different stages in aerobic respiration, including:
 - glycolysis in the cytoplasm
 - link reaction
 - Krebs cycle
 - oxidative phosphorylation in mitochondria.